

ATHARVA KIRKOLE

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EDUCATION

Accelerated path: completed B.S. in 3 years + M.S. in 1 year — both in Computer Science.

Michigan State University

M.S. Computer Science — GPA 3.93 | Teaching Assistant, Object-Oriented Programming

East Lansing, MI
Aug 2025 – May 2026

Michigan State University

B.S. Computer Science, Minor in Business (Honors College) — GPA 3.93, Honors, Tau Beta Pi

East Lansing, MI
Aug 2022 – Apr 2025

Relevant Graduate coursework: Parallel Programming (CSE 822), Big Data Analysis (Hadoop / MapReduce), High-Performance Computing, Pattern Recognition, Algorithms.

EXPERIENCE

Michigan State University IT Services

Software Developer

East Lansing, MI
Jun 2024 – Apr 2025

- Led end-to-end migration of the campus web-mapping platform off Google Maps onto a self-managed ESRI ArcGIS stack serving **100K+ daily API requests** across **~35K** monthly active users, rebuilding every map surface around custom geospatial layers and cutting mapping/licensing cost **~20%** (**≈\$45K/yr**).
- Owned geospatial data modeling for **~12K+** campus location entities/POIs (buildings, entrances, accessibility nodes); designed layer rendering and an automated pipeline ingesting live location signals to optimize turn-by-turn routing, cutting route-compute latency **~30%** (**~210ms** → **~145ms**).
- Modernized campus web services (PHP, JavaScript, Docker, custom student-data REST APIs); raised site accessibility from **80%** to **99.7%** (WCAG 2.1 / ADA / ARIA) across **~40** production surfaces.

Heard AI

Speech AI Research Assistant

East Lansing, MI
Mar 2026 – Present

- Contributing to a \$5M NSF Convergence Accelerator (Phase 2) project on accessible speech AI.
- Built a full-stack, real-time speech application (Node.js, React, AWS Cloudscape) on a Claude-based LLM framework serving **~300** concurrent sessions at **<400ms** end-to-end latency.
- Developed a personalized per-user fine-tuning pipeline (**~4 min/user**), improving recognition accuracy **23%** (WER **~14%** → **~10.8%**) via rapid adaptation.

Amazon

SDE Intern, University Capstone

East Lansing, MI
Jan 2025 – May 2025

- Built an AI-powered semantic code-search system over **~1.2M+** internal architecture assets using AWS OpenSearch (Elasticsearch-compatible), Lambda, DynamoDB, and S3.
- Designed a RAG-style retrieval pipeline (dynamically chunked vector embeddings via Bedrock + SageMaker, indexed in OpenSearch) at **~90ms** p95; tuned indexing/ranking to lift top-k relevance **~18%** (NDCG@10) and added vector-similarity employee recommendations over GitHub signals.

Thompson Maize Lab

Machine Learning Research Intern

East Lansing, MI
Aug 2023 – May 2024

- Built and evaluated Random Forest / ML models (TensorFlow, Python, R) for crop-yield prediction on 120+ dim datasets (**~72%** accuracy); presented at NAPPN Conference, Purdue, with a published abstract ([link](#)).

PROJECTS

FitTrack — Wearable Time-Series Analytics & Health Prediction (Python / TensorFlow)

2025

- Built a fitness app that ingests **live time-series data** from wearable devices (heart rate, accelerometer, motion signals), visualizing streaming metrics on a real-time dashboard at **~50Hz** sampling.
- Trained ML models predicting **age, gender, BMI, and body-fat %** from sensor signals — **~88%** accuracy, **~3.5% MAE** on body-fat regression across **~1.5M** sample windows.

NanoGPT Attention Kernel Optimization (CUDA / C++17 / OpenMP)

Jan 2026

- Implemented Transformer multi-head attention in C++17 over [B, H, N, D] tensors with a 4D accessor collapsing index/stride arithmetic; row-tiled Q-KT and P-V matmuls to keep K/V panels resident in L1/L2.
- Parallelized across (batch, head) with OpenMP, vectorized with AVX2 + FMA, and fused matmul-softmax-matmul for a 68× speedup; built a CUDA variant with cooperative shared-memory tiling and per-thread online-softmax state.

Tensor-Core GEMM (CUDA) — CSE 822 Parallel Programming

Jan 2026

- Implemented a TF32 tensor-core GEMM using `mma.sync.m16n8k8` PTX intrinsics, with float4-vectorized coalesced HBM loads and padded shared memory to eliminate bank conflicts; benchmarked across $M \in [1, 3072]$ at $K = N = 3072$.

TECHNICAL SKILLS

Maps & Geospatial: ESRI ArcGIS, ArcGIS Portal, Google Maps API, Leaflet.js, GIS data modeling, POI/entity modeling, A* routing

Search & Big Data: Hadoop / MapReduce, Spark/PySpark, OpenSearch / Elasticsearch, vector embeddings (Bedrock / SageMaker), RAG, time-series analytics, DynamoDB, MongoDB, S3

Languages & ML/Web: C++17, C#, Python, JavaScript, SQL, R, PHP; PyTorch, TensorFlow, React, Node.js, Flask, Django, .NET, REST APIs, Docker, AWS Lambda; CUDA / OpenMP / SIMD

CERTIFICATIONS

Graduate Certificate in High-Performance Computing, MSU (Expected May 2026) • CS50x, HarvardX (Verified).